industrial plant), long-range technologies become essential. Table 3.4 provides a comparison of wireless communication technologies that have ranges in excess of 1000 meters. Each of these technologies makes different tradeoffs to achieve their extended range. LPWAN devices have a range of kilometers on battery power, but data rates are measured in bits per second. Cellular operates on licensed frequencies with guaranteed service levels but requires monthly subscription fees, while low-earth orbit satellite services like Starlink have made communication practical in the middle of the ocean, though at a premium price.

Table 3.4 Popular long-range wireless communication technologies compared on data rate, range, battery life and cost model.

Technology	Data rate	Typical range	Battery life	Cost model	Home use case
LPWAN (LoRaWAN)	0.3-50kbps	2-15km urban	5-10 years	One-time hardware	Selected for indoor and outdoor sensors
Cellular	100kbps-50Mbps	National coverage	6-12 months	Monthly fees	Too expensive for my needs
Satellite	25Mbps+	Global coverage	Hours	High monthly fees	Unnecessary for a suburban home

3.3.4 Choosing the right technology

In my home deployment, the choice became clear once I mapped out sensor locations. Indoor sensors near power outlets could leverage my existing WiFi network; there was no need for new infrastructure where I had solid coverage. The electricity meter presented a unique challenge since I could not find an affordable LoRaWAN sensor. Instead, I used a low cost pulse counter that syncs with a smartphone app via BLE, making it the pragmatic choice despite requiring manual data export to transfer the data to my collection hub. Given its popularity in home automation, I seriously evaluated Zigbee for its mesh networking capabilities, but ultimately decided that deploying a separate Zigbee network made little sense when LoRaWAN could handle all my battery-powered sensors in the required range with a single gateway. Figure